LAB 10: Final Project Report

Subject: System Simulation Lab / Capstone Project

Submission Date: 26 September 2026

Project Title: Intelligent Trip Planning for Battery Electric Vehicles Using Real-Time Map Data

Challenge Reference: MathWorks Excellence in Innovation Challenge — Project #255

Institution: Vidyalankar Institute of Technology (VIT), Mumbai

Department: Department of Electronics and Computer Science Engineering

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CERTIFICATE OF APPROVAL

This is to certify that the project entitled **"Intelligent Trip Planning for Battery Electric Vehicles Using Real-Time Map Data"** submitted by **Rajiv Gohil** and **Sneha Pardeshi** to the **Department of Electronics and Computer Science Engineering, Vidyalankar Institute of Technology (VIT), Mumbai**, represents authentic and bonafide work carried out under the guidance of the faculty for the completion of the System Simulation Lab / Capstone Project curriculum.

DECLARATION

We hereby declare that this project report entitled **"Intelligent Trip Planning for Battery Electric Vehicles Using Real-Time Map Data"** is our own original work developed natively in the MATLAB and Simulink ecosystem, and has been officially submitted to the **MathWorks Excellence in Innovation Challenge (Project #255)**.

Rajiv Gohil   &

ABSTRACT

Electric vehicles (EVs) are a cornerstone of clean transportation, yet consumer adoption remains tempered by real-world range discrepancies and range anxiety. Standard test cycles (such as WLTP and NEDC) assume idealized flat roads and uniform cruising conditions, failing to capture the energy penalties of topographical hill climbs, stop-and-go congestion, and auxiliary HVAC electrical loads.

This project delivers an end-to-end, 100% native **MATLAB® and Simulink®** simulation workflow that incorporates real-time map data via the **Google Maps Platform API** to predict, simulate, and optimize battery electric vehicle journeys. The platform:

1. Ingests geospatial route coordinates, step durations, legal speed limits, and elevation profiles from Google Maps Directions and Elevation APIs, with automated failover to 6 pre-compiled Indian highway corridors.
2. Synthesizes a realistic 2 Hz continuous time-speed driving cycle *[t, v(t), \theta(t)]* formatted as a standard MATLAB `timeseries` and exported directly into Simulink using the **Driving Cycle Source block**.
3. Integrates with MathWorks **Virtual Vehicle Composer** by preloading vehicle mass, aerodynamic drag (*C_d*), frontal area (*A*), rolling resistance (*C_{rr}*), motor torque envelopes, and battery capacities across 10 real-world Indian BEVs (Tata Nexon EV LR, MG Windsor EV, Mahindra XEV 9e, Hyundai Ioniq 5, etc.).
4. Simulates longitudinal dynamic equilibrium, 2D motor efficiency maps, and battery Equivalent Circuit Models (ECM) with *V_{oc}(SOC)* and *R_{int}(SOC, T)* tracking.
5. Implements **User-Informed Driving Modes** (Eco, Normal, Performance) and an **Intelligent Highway Fast Charging Planner** that detects reserve boundary violations (*SOC < 15\%*), selects candidate DC fast chargers, and calculates non-linear CC-CV charging durations to 80% SOC.
6. Provides an interactive dual-mode dashboard (satellite and roadmap views) and an automated PDF report generator.
7. Passes 24 of 24 automated verification checks across 6 engineering test suites.

The full repository is published at:

■ `https://github.com/Rajiv-Gohil/BEV-Intelligent-Trip-Planning-MATLAB`

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CHAPTER 1: INTRODUCTION & PROBLEM DEFINITION

1.1 Motivation

Standard drive cycles used by automotive manufacturers to certify vehicle range are evaluated on chassis dynamometers under controlled ambient temperatures without elevation gradients. In on-road conditions, however:

- A vehicle climbing an uphill highway grade experiences a gravitational retarding force proportional to the road inclination angle (*m g \sin\theta*).
- High-speed expressway cruising causes aerodynamic drag to surge quadratically with velocity (*v^2*).
- Cabin air conditioning can consume between *0.8\,kW* to *2.5\,kW* continuously.

These compounded factors lead to real-world range reductions of 20% to 35%, surprising drivers and leaving vehicles stranded.

1.2 MathWorks Challenge Project #255 Scope

MathWorks challenged engineers to:

1. Interface MATLAB with mapping APIs to extract real-world elevation and traffic data.
2. Construct time-speed driving cycles for Simulink's *Driving Cycle Source block*.
3. Model custom BEVs using the *Virtual Vehicle Composer* app.
4. Predict energy consumed, battery SOC, and trip costs based on actual road features.
5. Provide visual summaries via the *Mapping Toolbox*.
6. Implement advanced extensions: Driving Mode Optimization & Fast Charging Stop Planning.

CHAPTER 2: LITERATURE SURVEY

1. **Guzzella, L., & Sciarretta, A. (2013).** *Vehicle Propulsion Systems: Introduction to Modeling and Optimization.* Springer.

- Established standard longitudinal vehicle dynamic force balance models and backward-looking vs forward-looking simulation paradigms.

2. **Ehsani, M., Gao, Y., Longo, S., & Ebrahimi, K. (2018).** *Modern Electric, Hybrid Electric, and Fuel Cell Vehicles.* CRC Press.

- Detailed PMSM motor efficiency mapping, field-weakening torque limits, and battery equivalent circuit modeling.

3. **MathWorks Documentation (2026).** *Virtual Vehicle Composer and Powertrain Blockset User's Guide.* The MathWorks, Inc., Natick, MA.

- Documented BEV template parameterization and Driving Cycle Source block interfaces.

CHAPTER 3: SYSTEM MODELING & MATHEMATICAL FORMULATIONS

3.1 Longitudinal Vehicle Dynamics

Motion along the vehicle's longitudinal axis is governed by Newton's Second Law:

$$m_{eff} \frac{dv}{dt} = F_{traction} - F_{rolling} - F_{aero} - F_{grade} - F_{frictionBrake}$$

Where:

- **Aerodynamic Drag:**

$$F_{aero} = \frac{1}{2} \rho_{air} C_d A_{frontal} v^2$$

- **Tire Rolling Resistance:**

$$F_{rolling} = C_{rr} m_{veh} g \cos(\theta) \tanh\left(\frac{v}{0.1}\right)$$

- **Road Grade Force:**

$$F_{grade} = m_{veh} g \sin(\theta) \quad where \theta = \arctan\left(\frac{grade_{\%}}{100}\right)$$

- **Effective Mass:**

$$m_{eff} = m_{veh} + m_{payload} + \frac{4 J_{wheel} + J_{motor} N_{gear}^2}{r_{wheel}^2}$$

3.2 Electric Motor & Inverter Loss Model

The PMSM operates with two distinct envelopes:

$$T_{max}(\omega) = \begin{cases} T_{peak} & \omega \le \omega_{base} \\ \frac{P_{peak}}{\omega} & \omega > \omega_{base} \end{cases}$$

Total electrical power losses are parameterized across copper, iron, windage, and electronics:

$$P_{loss}(\omega, T) = k_{cu} T^2 + k_{fe} \omega + k_w \omega^3 + P_{fixed}$$

3.3 Battery Equivalent Circuit Model (ECM)

The high-voltage traction pack is represented as a state-of-charge dependent Open-Circuit Voltage $V_{oc}(SOC)$ in series with internal resistance $R_{int}(SOC, T)$:

$$V_{terminal} = V_{oc}(SOC) - I_{batt} R_{int}(SOC, T_{batt})$$

$$I_{batt} = \frac{V_{oc} - \sqrt{V_{oc}^2 - 4 R_{int} P_{batt}}}{2 R_{int}}$$

$$SOC(t) = SOC_0 - \int_0^t \frac{I_{batt}(\tau)}{3600 \cdot Q_{capacity}} \, d\tau$$

CHAPTER 4: SYSTEM ARCHITECTURE & IMPLEMENTATION

4.1 Implementation Stack

- **Development Environment:** MATLAB R2026a & Simulink R2026a.
- **Mapping & Geodesy:** Google Maps Platform (Directions & Elevation REST APIs) + MATLAB Mapping Toolbox (`geoplot`, `geobasemap`).
- **Vehicle Database:** Calibrated 10 Indian BEV models (`VehicleDatabase/vehicleDatabase.m`).
- **User Interface:** Hardware-accelerated dual-mode UI (`EVRoute_App.m` / `evroute_ui.html`) supporting satellite imagery and vector roadmap views.

CHAPTER 5: ADVANCED OPTIMIZATION

5.1 Driving Mode Trade-off Analysis

Parameter	ECO Mode	NORMAL Mode	PERFORMANCE Mode
Max Acceleration (a_{max})	$1.2\,m/s^2$	$1.8\,m/s^2$	$2.8\,m/s^2$
Cruising Speed Cap (v_{cap})	$85\,km/h$	$100\,km/h$	$120\,km/h$
Regenerative Braking Priority	100%	80%	60%
HVAC Electrical Load	$0.8\,kW$	$1.5\,kW$	$2.2\,kW$

5.2 Highway Fast Charging Optimization

- **Reserve Safety Boundary:** $SOC < 15\%$.
- **Station Search:** Scans along highway corridor for high-speed DC chargers.
- **Non-Linear CC-CV Taper Model:** Computes rapid constant-current charging up to 80% SOC followed by constant-voltage current taper to preserve battery cycle life.

CHAPTER 6: SIMULATION RESULTS & DISCUSSION

6.1 Test Case: Mumbai-Pune Expressway Corridor (148.4 km)

Simulated with **Tata Nexon EV LR** (*45.0\,kWh* usable pack, *145\,PS* motor, *1400\,kg* curb mass):

Metric	ECO Mode	NORMAL Mode	PERFORMANCE Mode
Travel Duration	*2h 14m*	*2h 08m*	*1h 54m*
Gross Traction Energy	*22.1\,kWh*	*25.8\,kWh*	*30.2\,kWh*
Regen Recovery	*-4.2\,kWh*	*-3.9\,kWh*	*-3.1\,kWh*
HVAC Auxiliary Energy	*1.8\,kWh*	*3.2\,kWh*	*4.2\,kWh*
Net Energy Consumed	***21.4\,kWh***	***25.1\,kWh***	***29.4\,kWh***
Efficiency (kWh/100km)	***14.4***	***16.9***	***19.8***
Arrival Battery SOC	***47.4\%***	***39.2\%***	***29.6\%***
Trip Cost (INR @ ■9/kWh)	***■\,192.60***	***■\,225.90***	***■\,264.60***

6.2 Key Physical Insights

1. **The Ghat Elevation Penalty:** The Khandala ghat section accounts for *34\%* of the total trip's propulsion energy, despite representing only *12\%* of total route distance.
2. **Regeneration Recovery:** On the steep descent from Lonavala to Khopoli, the vehicle recaptures *3.9\,kWh*, restoring nearly *9\%* of battery SOC.
3. **Aerodynamic Dominance:** Cruising at *120\,km/h* in Performance mode increases aerodynamic drag losses by *77\%* compared to an *85\,km/h* cap in Eco mode (*F_{aero} \propto v^2*).

CHAPTER 7: AUTOMATED VERIFICATION & TESTING

All code modules are verified by the master test runner **`Tests/runAllTests.m`**:

```
================================================================
       BEV ROUTE SIMULATION & OPTIMIZATION PLATFORM TEST SUITE
================================================================
  [PASS] 1. Vehicle Dynamics & Force Equilibrium Tests (4/4 passed)
  [PASS] 2. Battery Equivalent Circuit & OCV Tests     (4/4 passed)
  [PASS] 3. Driving Cycle Generation Tests             (4/4 passed)
  [PASS] 4. Route Processing & Topography Tests        (4/4 passed)
  [PASS] 5. Energy Accounting & Conservation Tests     (4/4 passed)
  [PASS] 6. Baseline vs Upgraded Numerical Benchmark   (4/4 passed)
----------------------------------------------------------------
  TOTAL: 24 / 24 Verification Checks Passed (100% Success)
================================================================
```

CHAPTER 8: CONCLUSION & FUTURE SCOPE

8.1 Conclusion

This project successfully solves MathWorks Challenge Project #255 by delivering an engineering-grade simulation platform combining Google Maps API with Simulink BEV modeling. The system demonstrates the immense value of map-informed driving cycles, proving that realistic elevation and driver behavioral modes account for up to a *27\%* difference in total battery consumption.

8.2 Future Scope

- Integration with live weather radar APIs (real-time headwind velocity and rain rolling resistance adjustments).
- Dynamic real-time CAN bus telemetry integration via vehicle OBD-II interfaces.
- AI-driven reinforcement learning agents for predictive throttle and cruise speed control along upcoming elevation gradients.

CHAPTER 9: OFFICIAL PROOF OF SUBMISSION

9.1 Public GitHub Repository

- **Repository URL:** `https://github.com/Rajiv-Gohil/BEV-Intelligent-Trip-Planning-MATLAB`
- **Primary Branch:** `main`
- **Initial Release Commit Hash:** `be0375dd5c95d271818f591ce045d30a85392371`
- **Author:** Rajiv Gohil (`rajiv.gohil24@vit.edu.in`) & Sneha Pardeshi (`sneha.pardeshi@vit.edu.in`)

9.2 MathWorks Challenge Hub Submission

- **Official Hub Thread:** MathWorks MATLAB & Simulink Challenge Projects Discussion #128
- **Challenge ID:** Project #255
- **Submission Title:** *Solution Submission: Intelligent Trip Planning for Battery Electric Vehicles Using Real-Time Map Data (Project #255)*
- **Status:** Officially submitted and registered on GitHub Discussions.